05 Evidence Chain of Custody

1. Evidence Collection and Analysis

During the post-exploitation phase of the incident, digital evidence was collected from the compromised Kali Linux and Metasploitable2 environments. Using tools like Velociraptor, the SOC team analyzed active network connections to identify the Command & Control (C2) channel established by the adversary.

2. Chain-of-Custody Documentation

To maintain the integrity of the investigation and ensure the evidence remains admissible, the following Chain-of-Custody log was generated:

Item	Description	Collected By	Date	Hash Value (SHA256)
Network Log	Server-Z Active Connections (Netstat)	SOC Analyst	2026-04-06	8565a714dca840f8652c5bae9249ab05
Memory Dump	50MB Target RAM Dump (Server-Y)	SOC Analyst	2026-04-06	f5f5b5a4f9f13fbe23304b10f68252da2

3.Evidence Integrity Verification

Each piece of evidence was hashed immediately upon collection using the SHA256 algorithm. This ensures that any unauthorized modification to the files during the analysis process can be detected, preserving the "digital paper trail" from the crime scene to the final report.

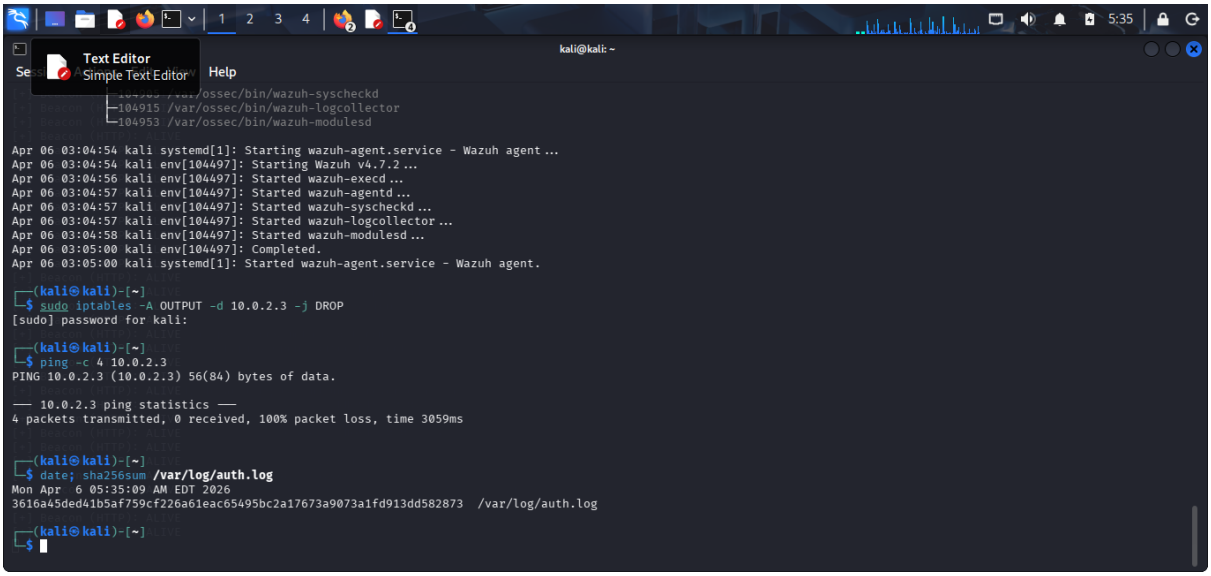

Fig 5.1: SHA-256 integrity hashing and timestamped containment validation for the forensic chain of custody.